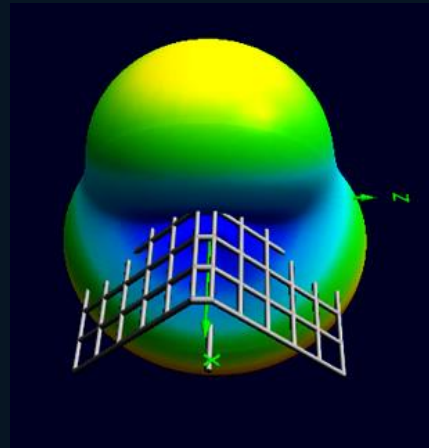
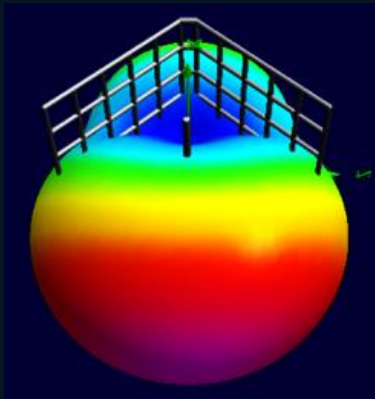
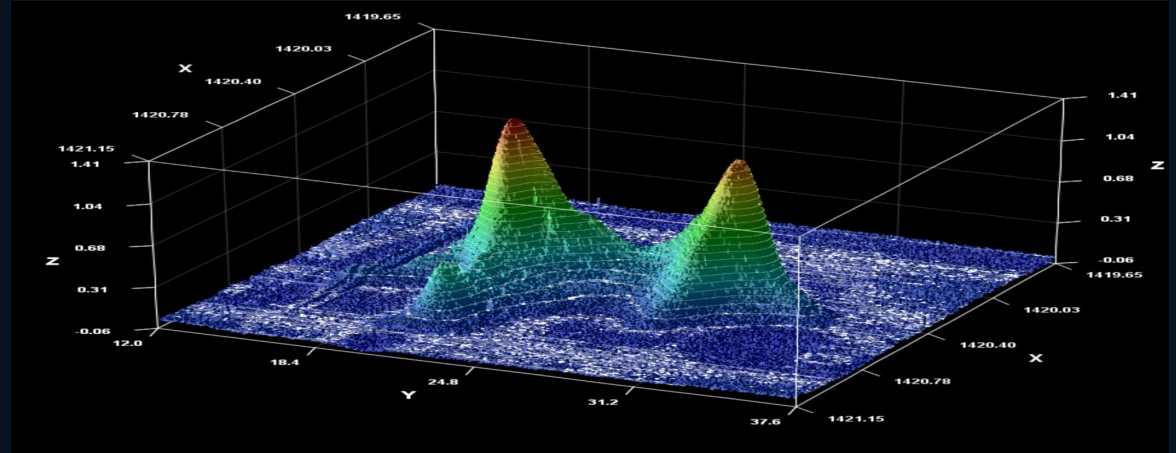
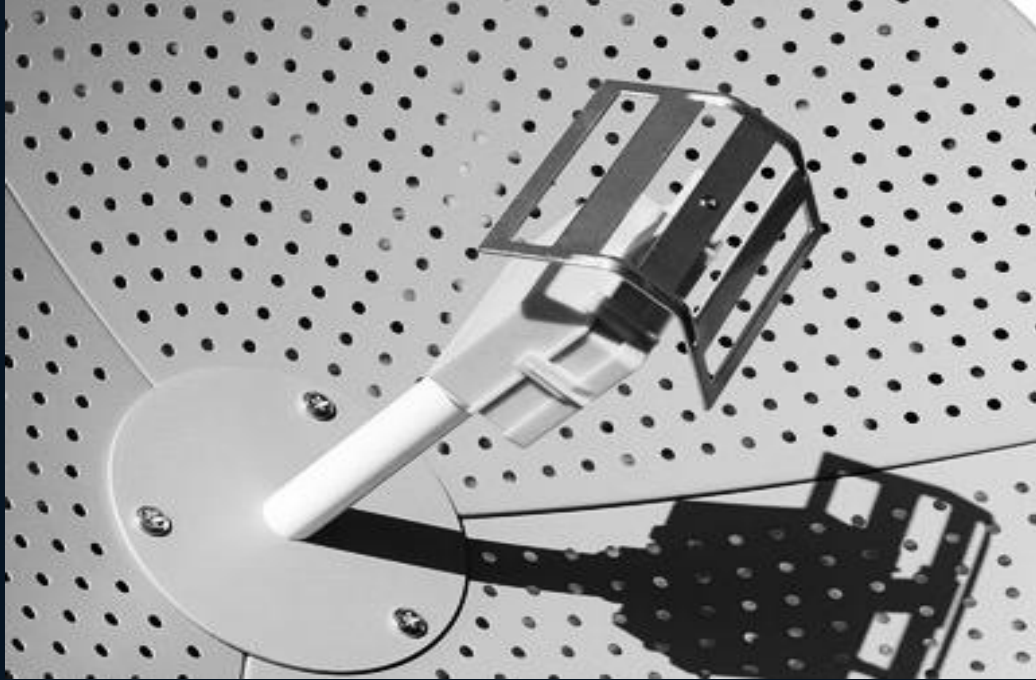
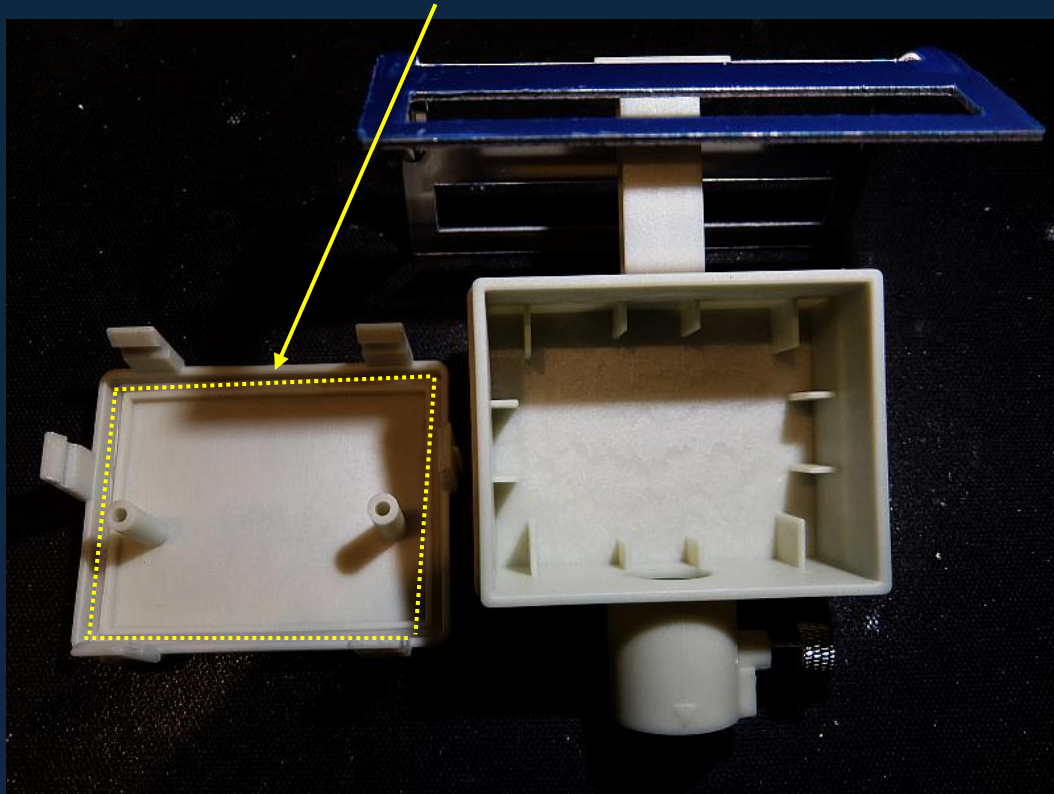


# *Kraken Discovery Dish 1420 MHz Feed Evaluation using a 1m WiFi Parabolic Reflector*

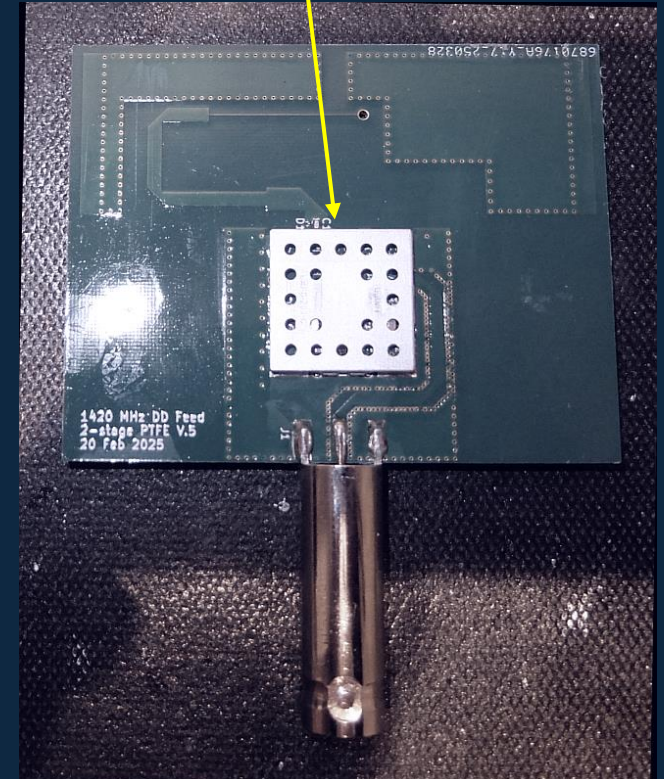
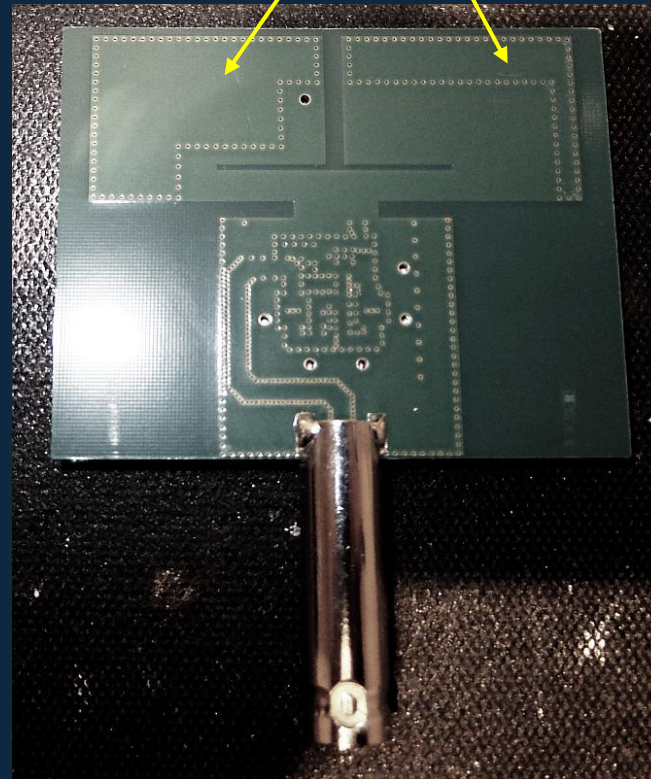


## Discovery Dish 1420 MHz Feed Evaluation Construction

Box has weather-seal around perimeter



Antenna is closely coupled to LNA  
to Minimize ( loss ) Noise Figure



# Kraken Discovery Dish 1420 MHz Feed Evaluation LNA

## Qorvo QPL9547 0.1GHz to 6.0GHz Ultra Low-Noise LNA

Qorvo QPL9547 0.1GHz to 6.0GHz Ultra-Low-Noise LNA is a high-linearity, ultra-low noise amplifier optimized for 5G Massive MIMO (Multiple Input Multiple Output) applications. At 1.9GHz, the amplifier typically provides 19.5dB gain, +39dBm OIP3 at a 65mA bias setting, and a noise figure of 0.3dB.

### Key Features

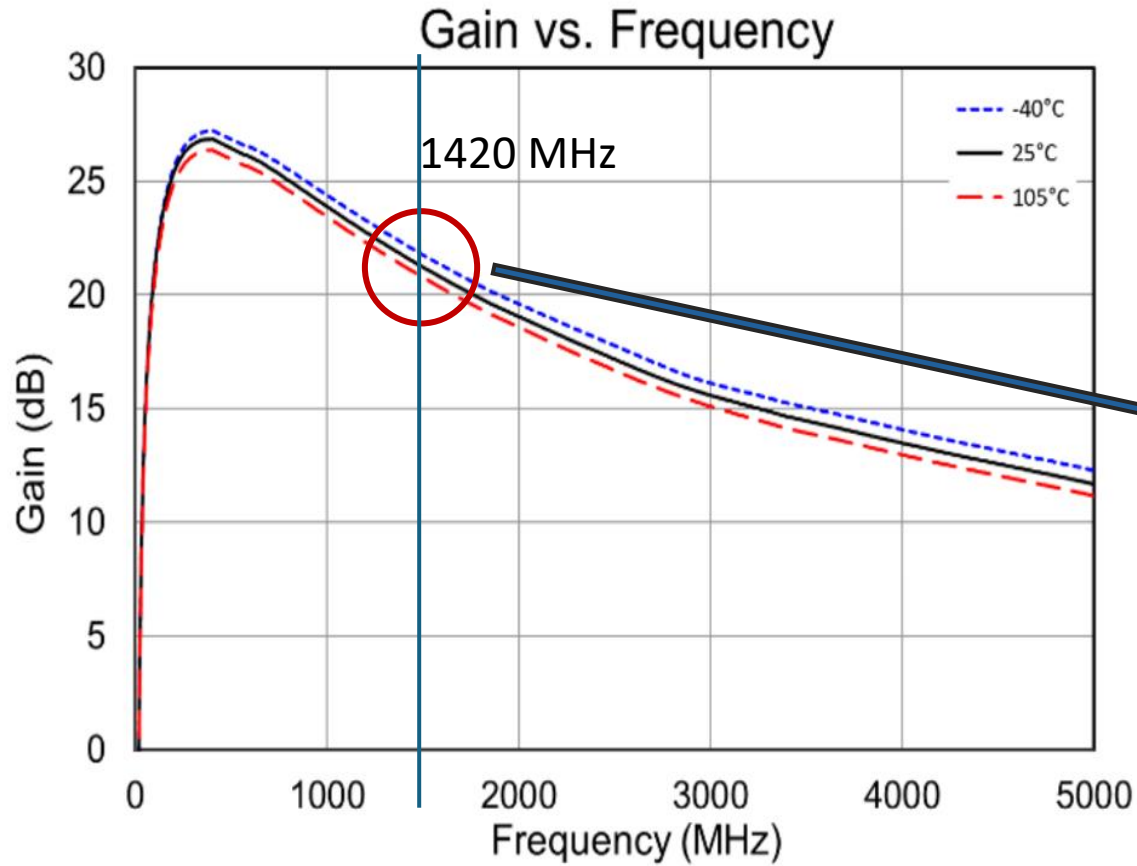
- 0.1-6 GHz Operational Bandwidth
- **Wide bandwidth: operates from 0.1GHz to 6 GHz**
- **Ultra-low noise: 0.3 db NF @ 1.9 GHz**
- **Gain: 19.5 dB typ.**
- **OIP3: +39 dBm typ.**

Qorvo QPL9547 0.1GHz to 6.0GHz Ultra Low-Noise LNA



# Kraken Discovery Dish 1420 MHz Feed Evaluation LNA

## Qorvo QPL 9547 LNA Information Gain vs Temperature Stability



### Gain vs. Frequency

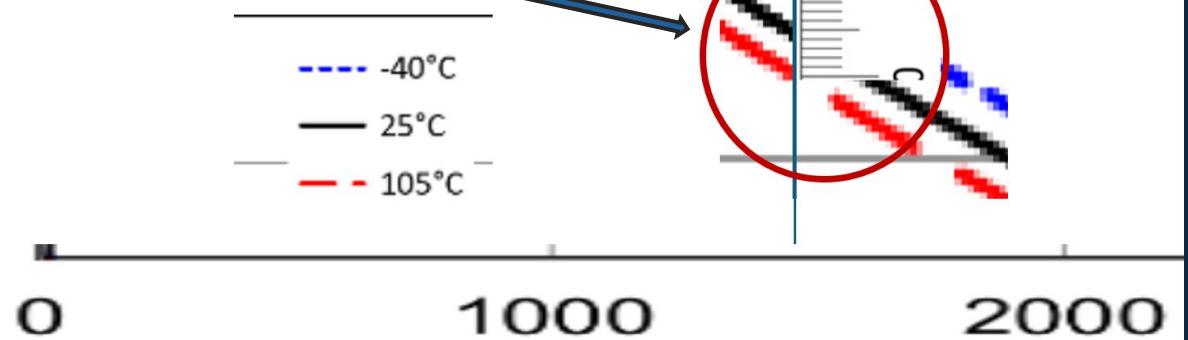
QORVO®

QPL9547  
Ultra Low-Noise Amplifier

#### Gain vs Temperature

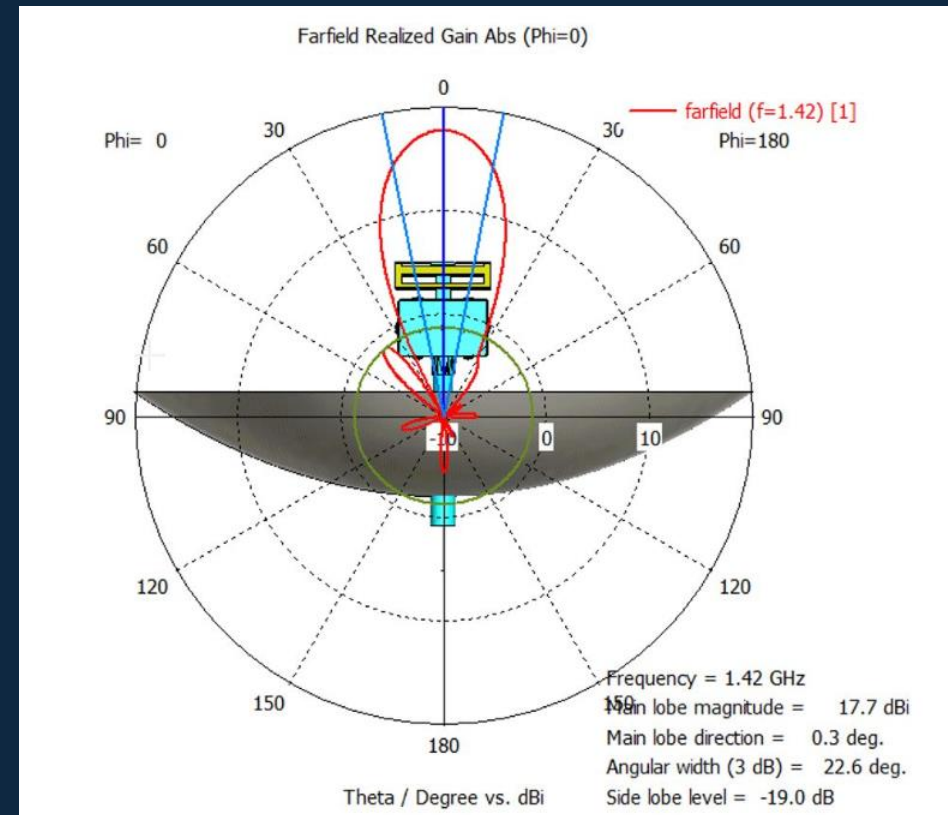
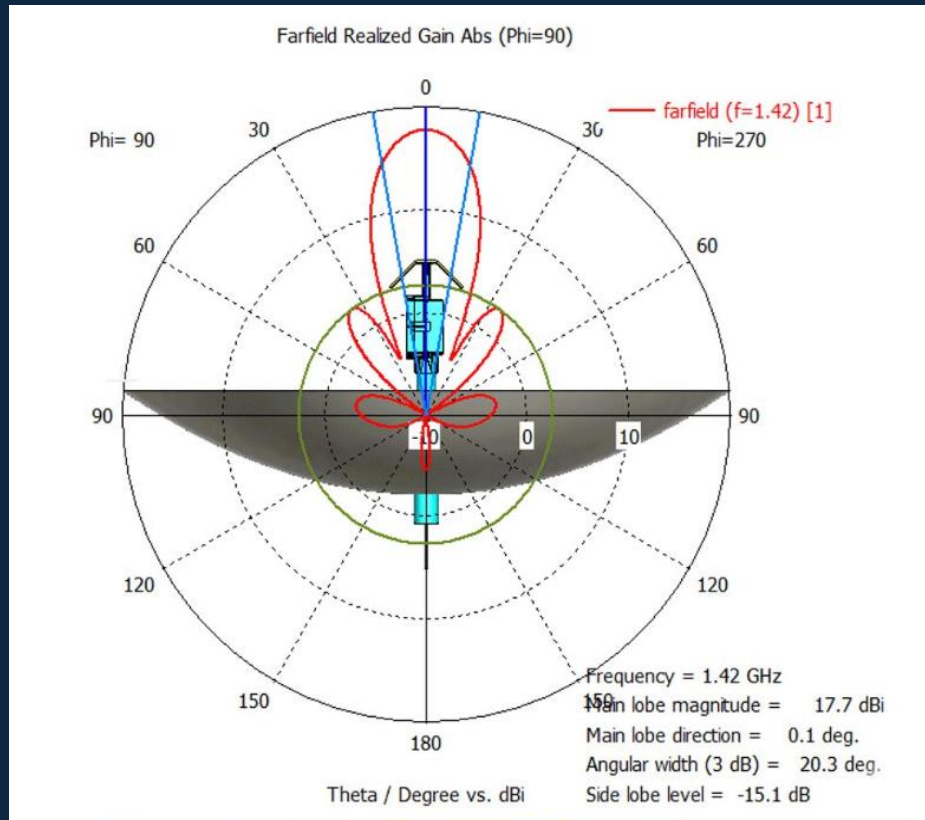
1.10 dB / 145° C = -0.075 dB / +10° C

0.45 dB / 80° C = -0.056 dB / +10° C



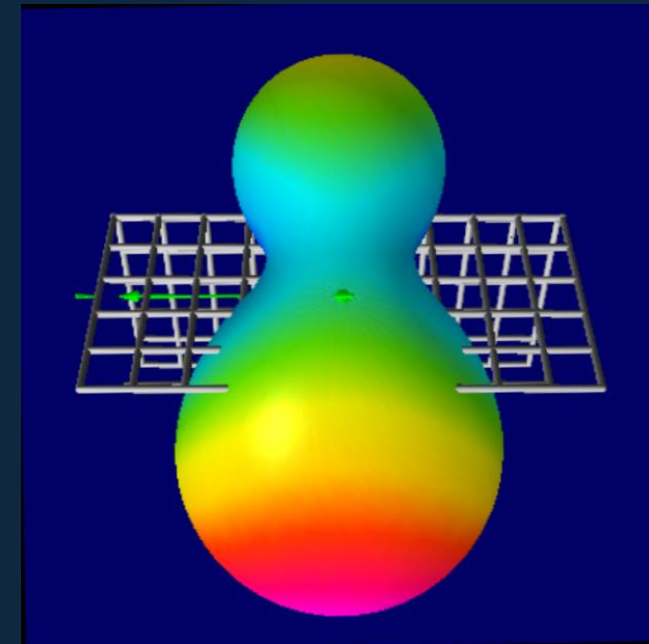
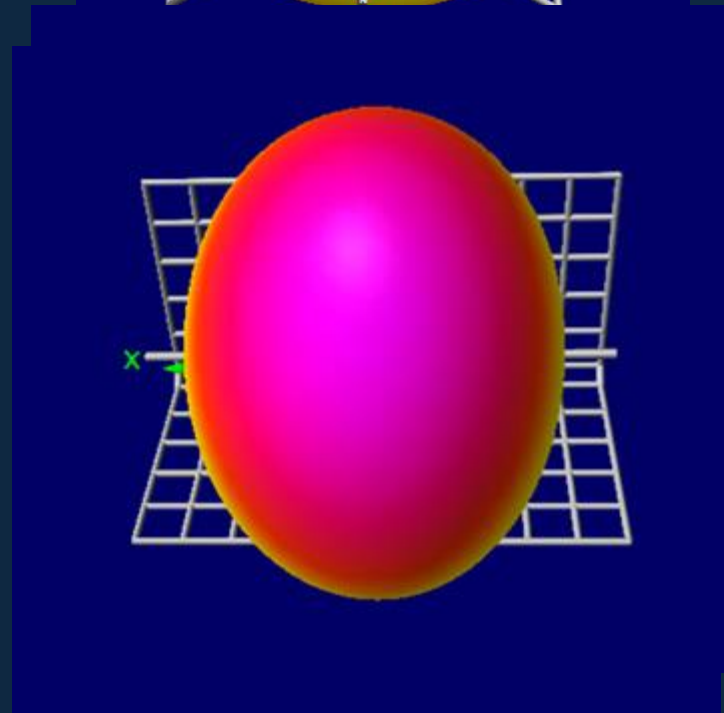
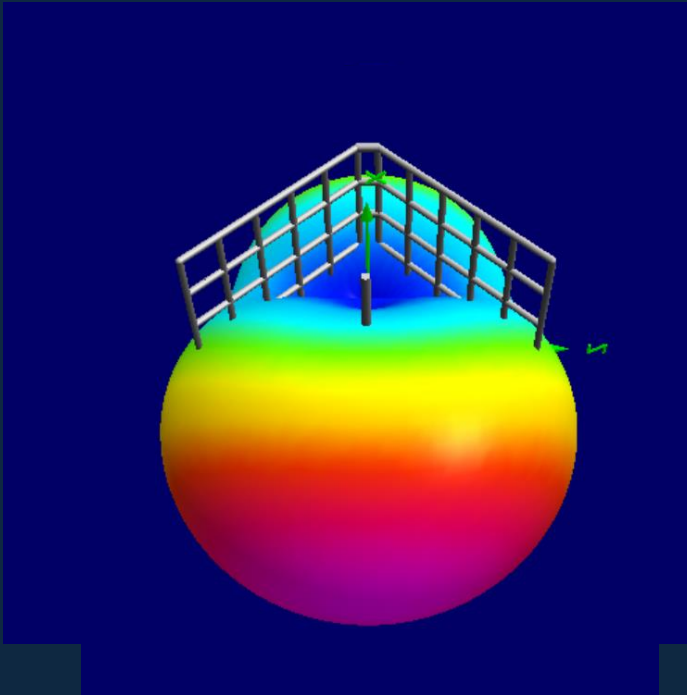
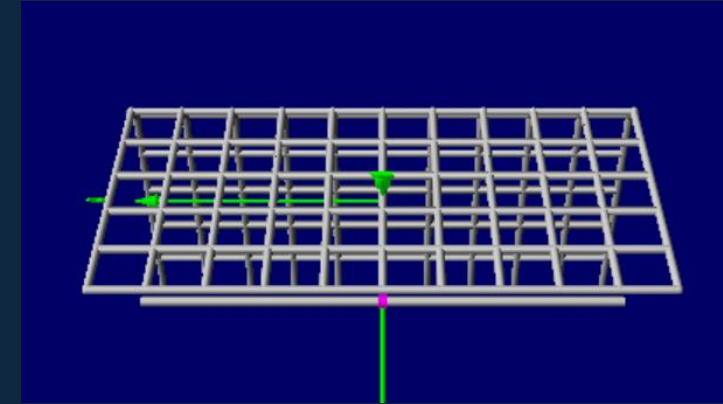
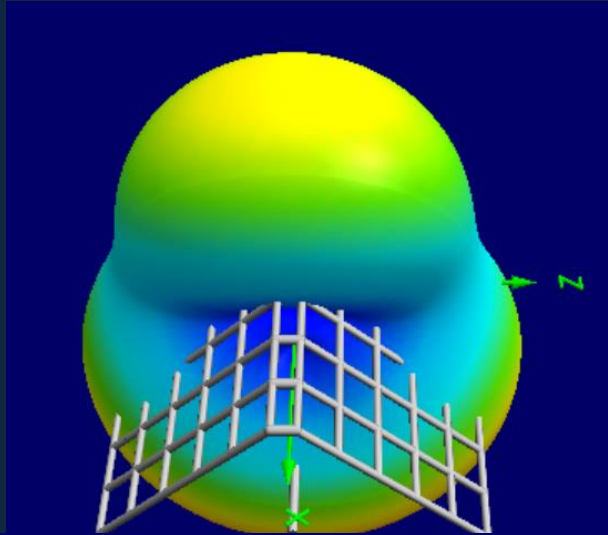
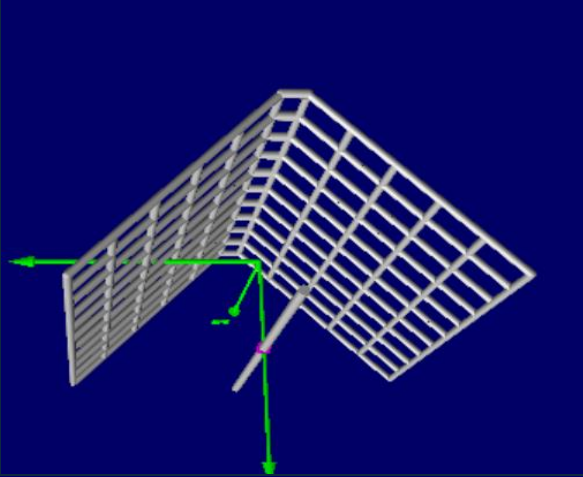
# Kraken Discovery Dish 1420 MHz Feed Evaluation

## Kraken information : Antenna Beam Patterns



**Kraken Discovery Dish 1420 MHz Feed Evaluation**  
**4NEC2 Numerical Electromagnetics Code Dipole/Reflector Analysis**  
*The actual antenna is more complex than this simple dipole .*

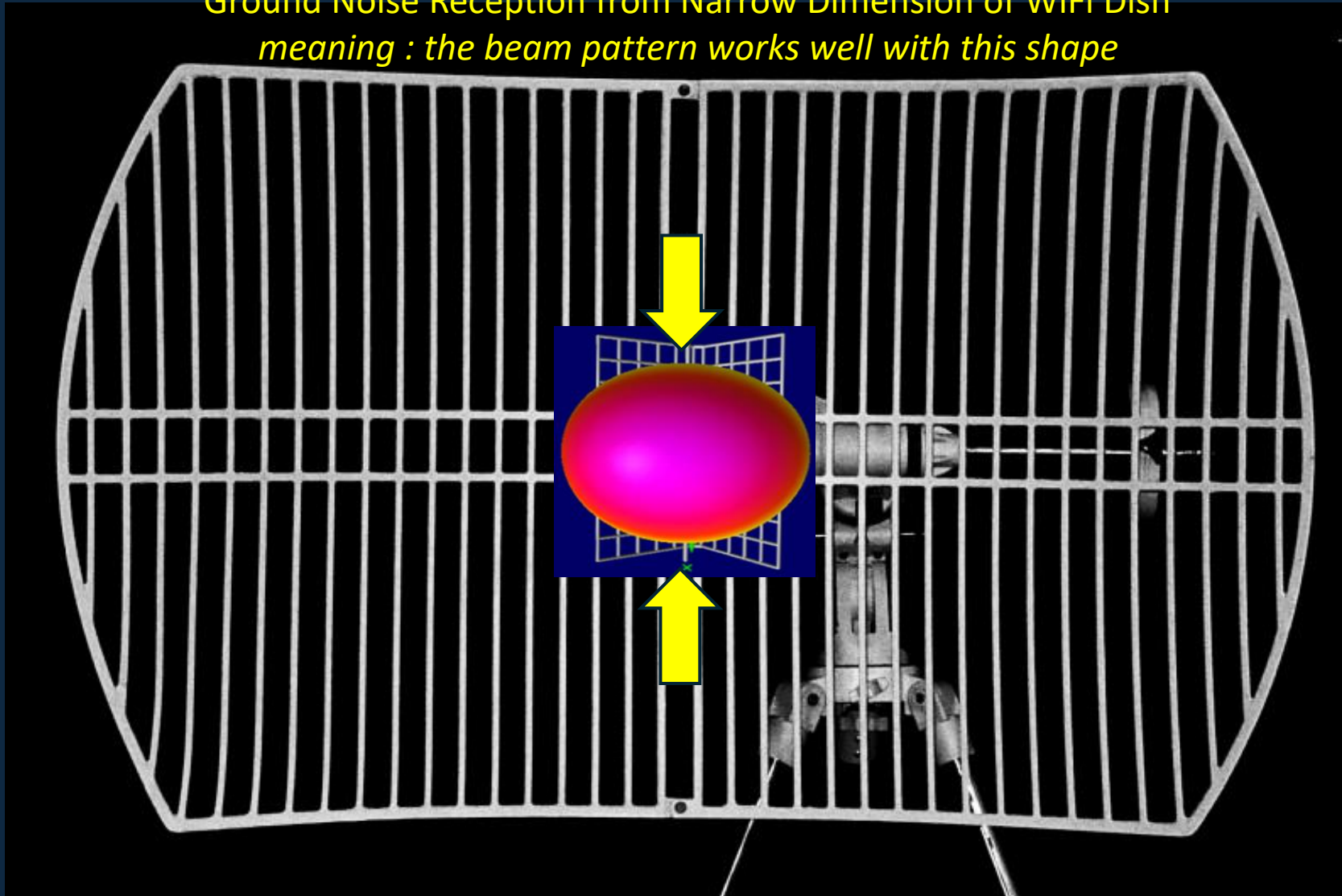
*alex petit*



**Kraken Discovery Dish 1420 MHz Feed Evaluation**  
**4NEC2 Numerical Electromagnetics Code Dipole/Reflector Analysis**  
*The actual antenna is more complex than this simple dipole .*

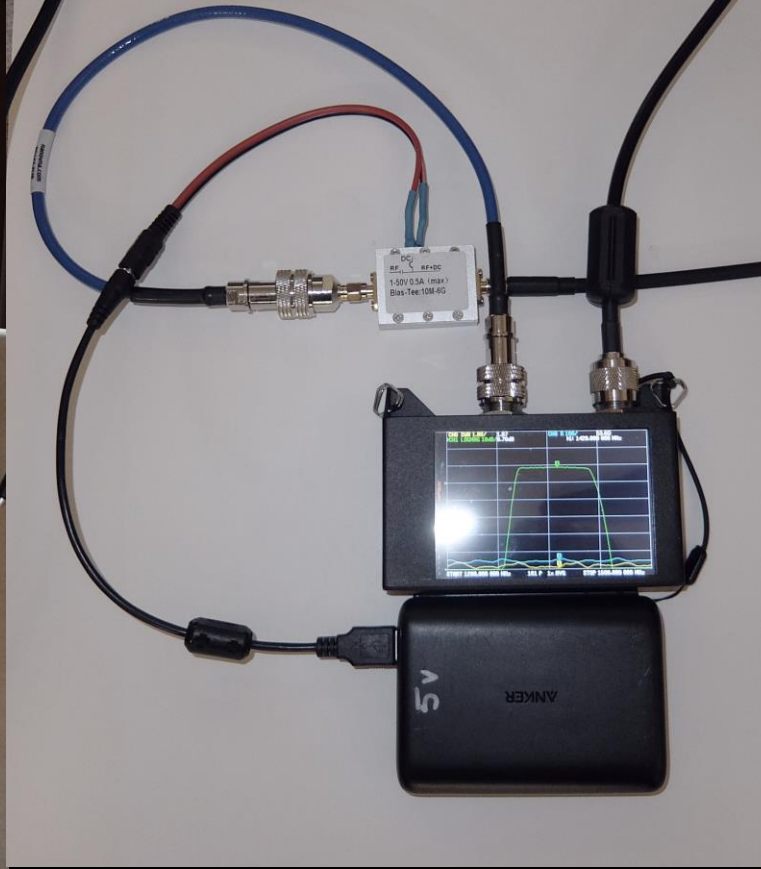
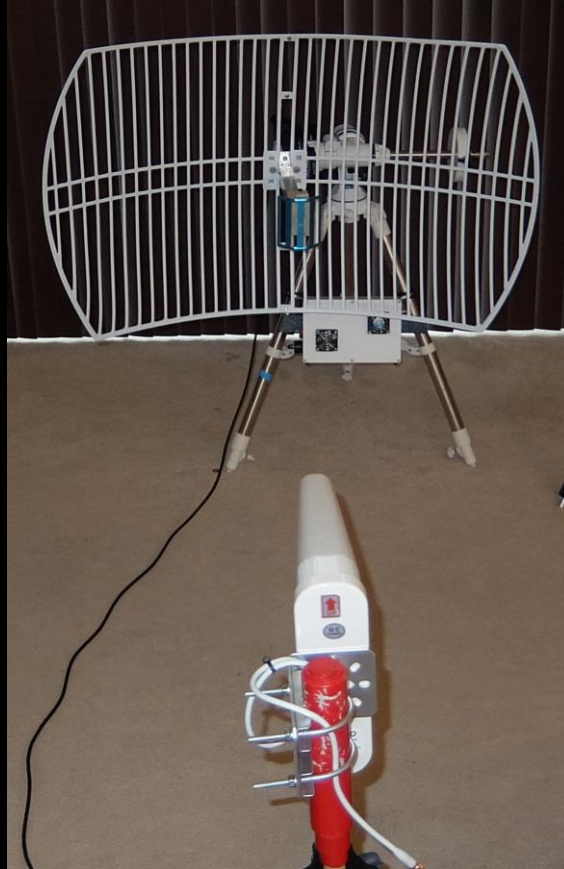
*alex petit*

Narrow Out-of-Axis Beam Pattern Minimizes  
Ground Noise Reception from Narrow Dimension of WiFi Dish  
*meaning : the beam pattern works well with this shape*



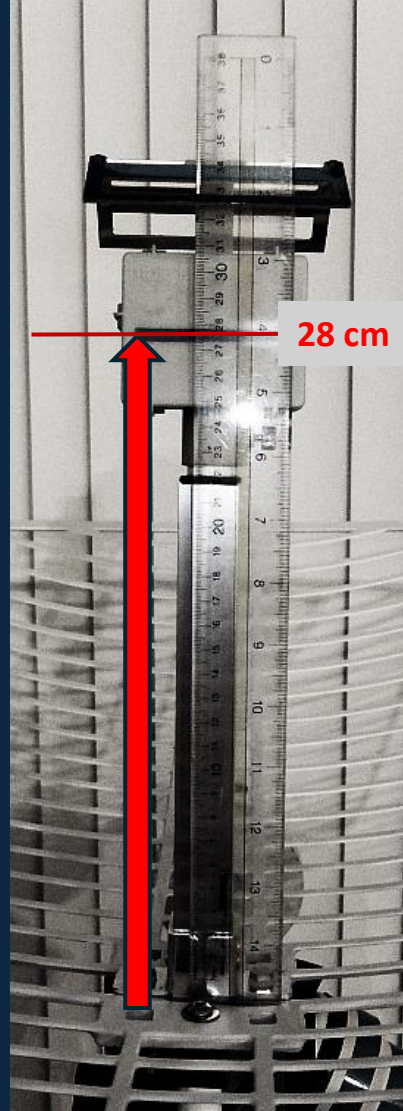
# Kraken Discovery Dish 1420 MHz Feed Evaluation

## Band Pass Filter Characteristics : 1380 – 1490 MHz

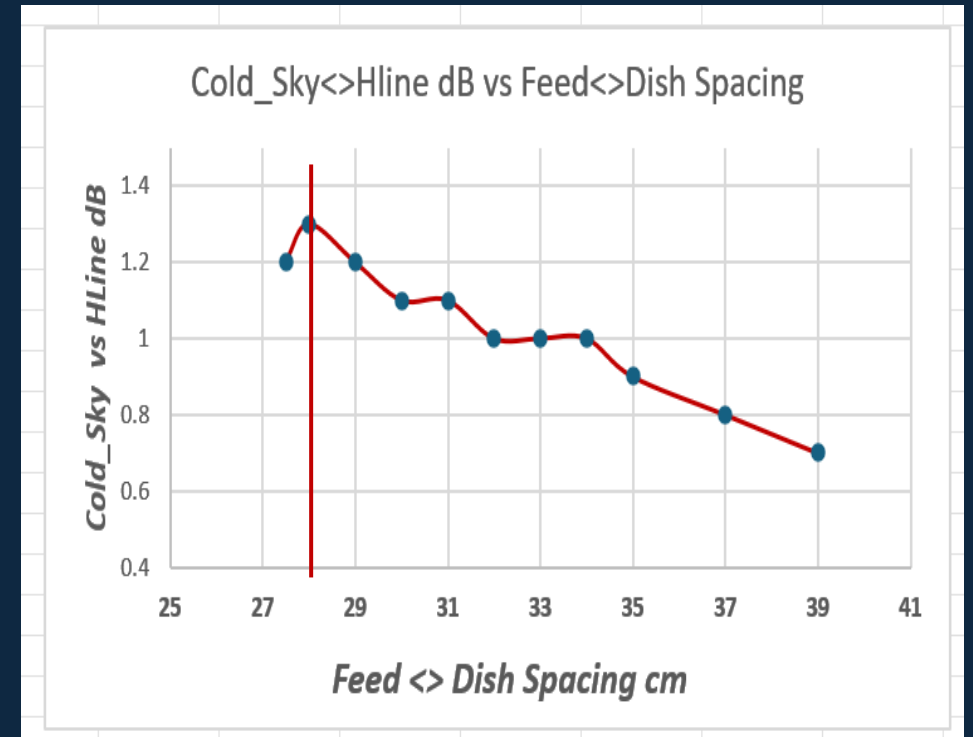


# Kraken Discovery Dish 1420 MHz Feed Evaluation

## WiFi Dish<>Feed Spacing Optimization

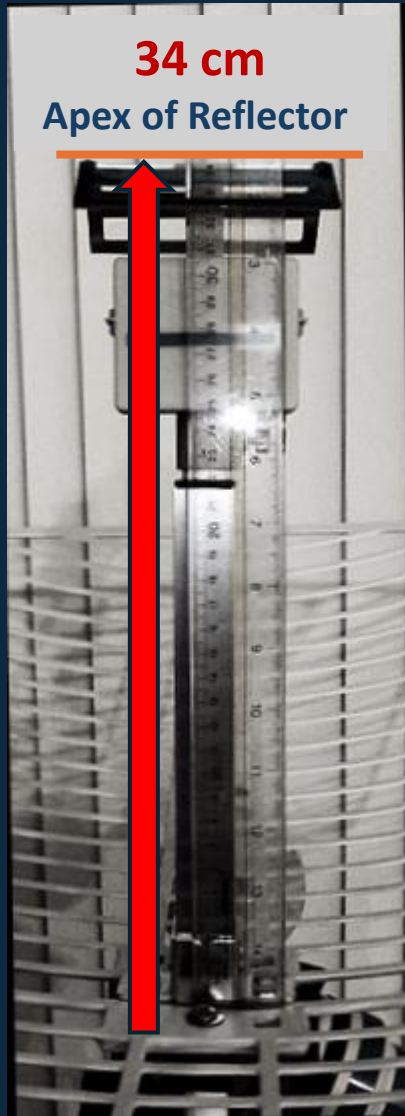


Nooelec 1m WiFi Grid Dish  
Discovery Dish 1420 MHz Dipole Feed  
Discovery Dish H1 internal LNA



# Kraken Discovery Dish 1420 MHz Feed Evaluation

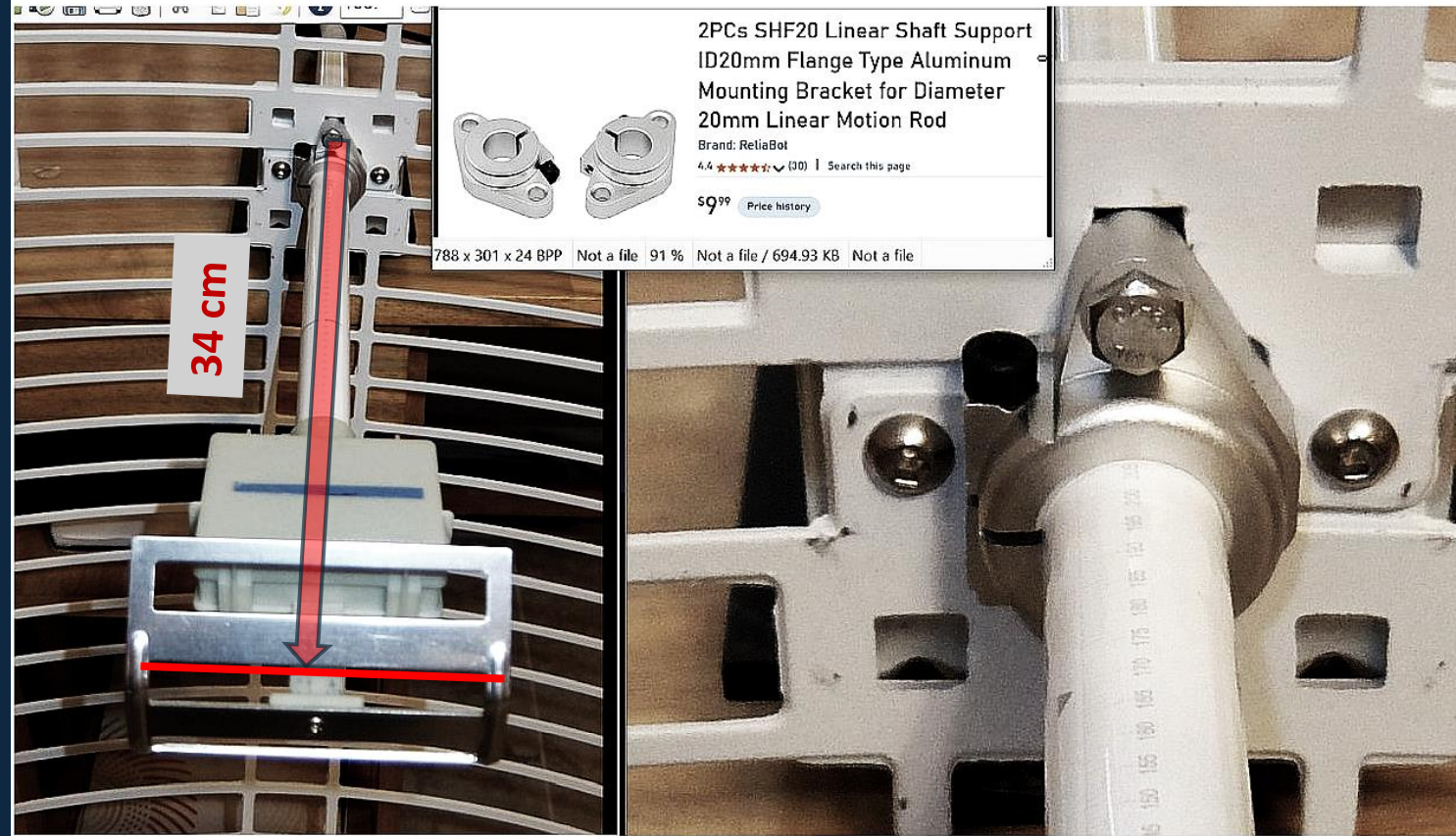
## WiFi Dish<>Feed Attachment Flange



SHF20 Linear Shaft Support  
ID 20mm Flange Type  
Aluminum Mounting Bracket for  
20mm Diameter Linear Motion Rod



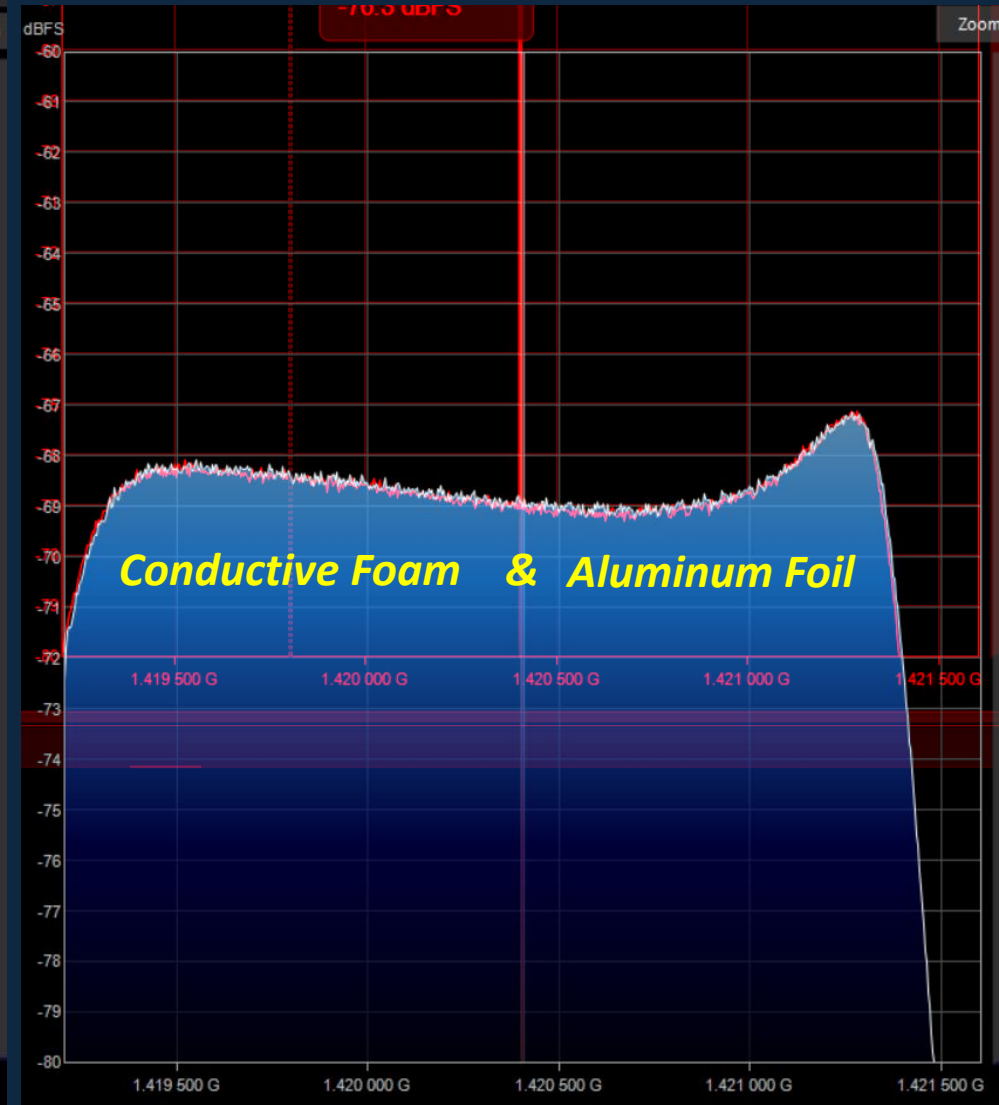
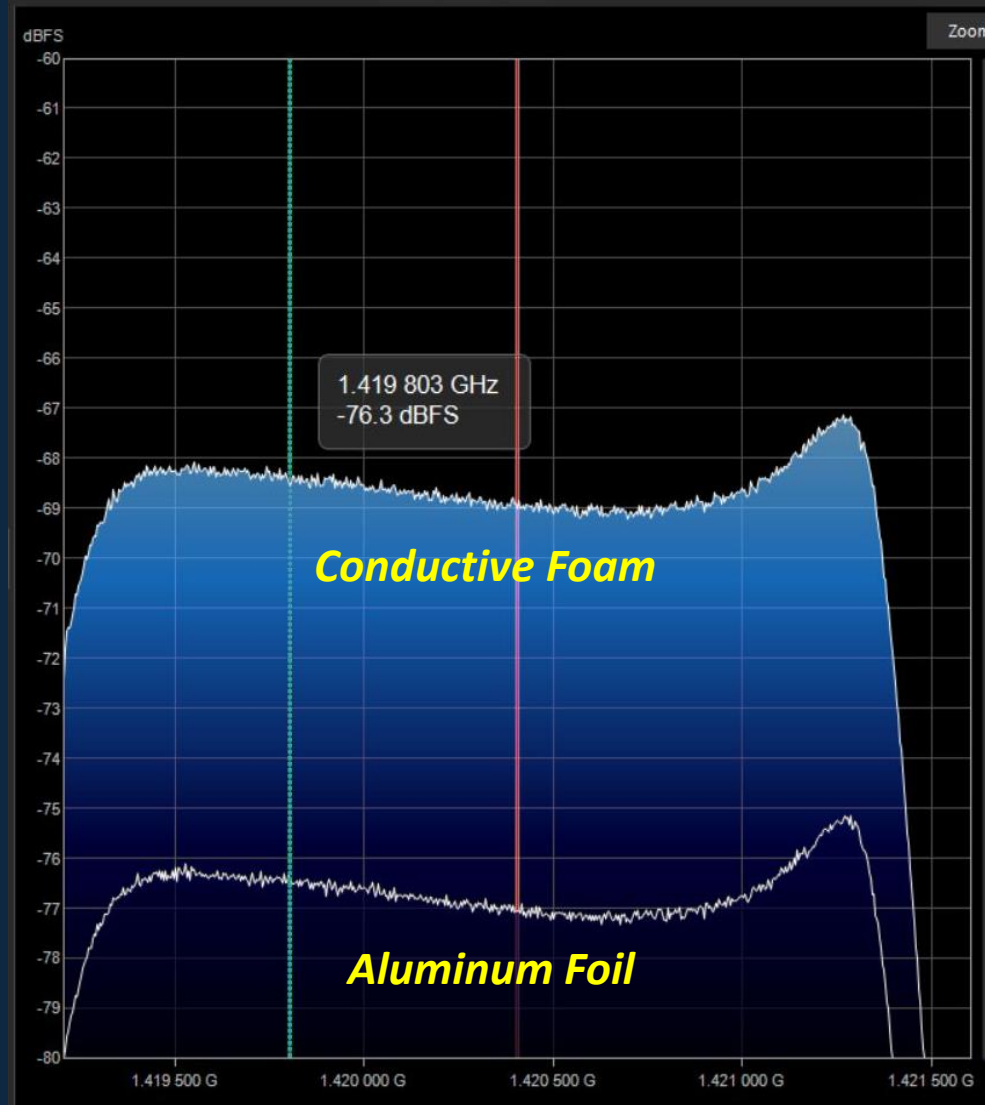
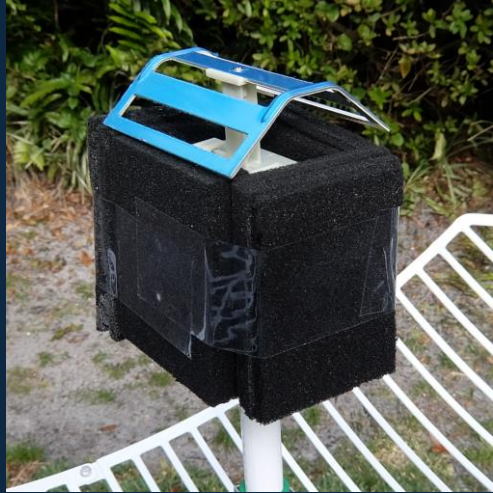
Its a perfect fit between the Flange Adapter & Feed Support Tube



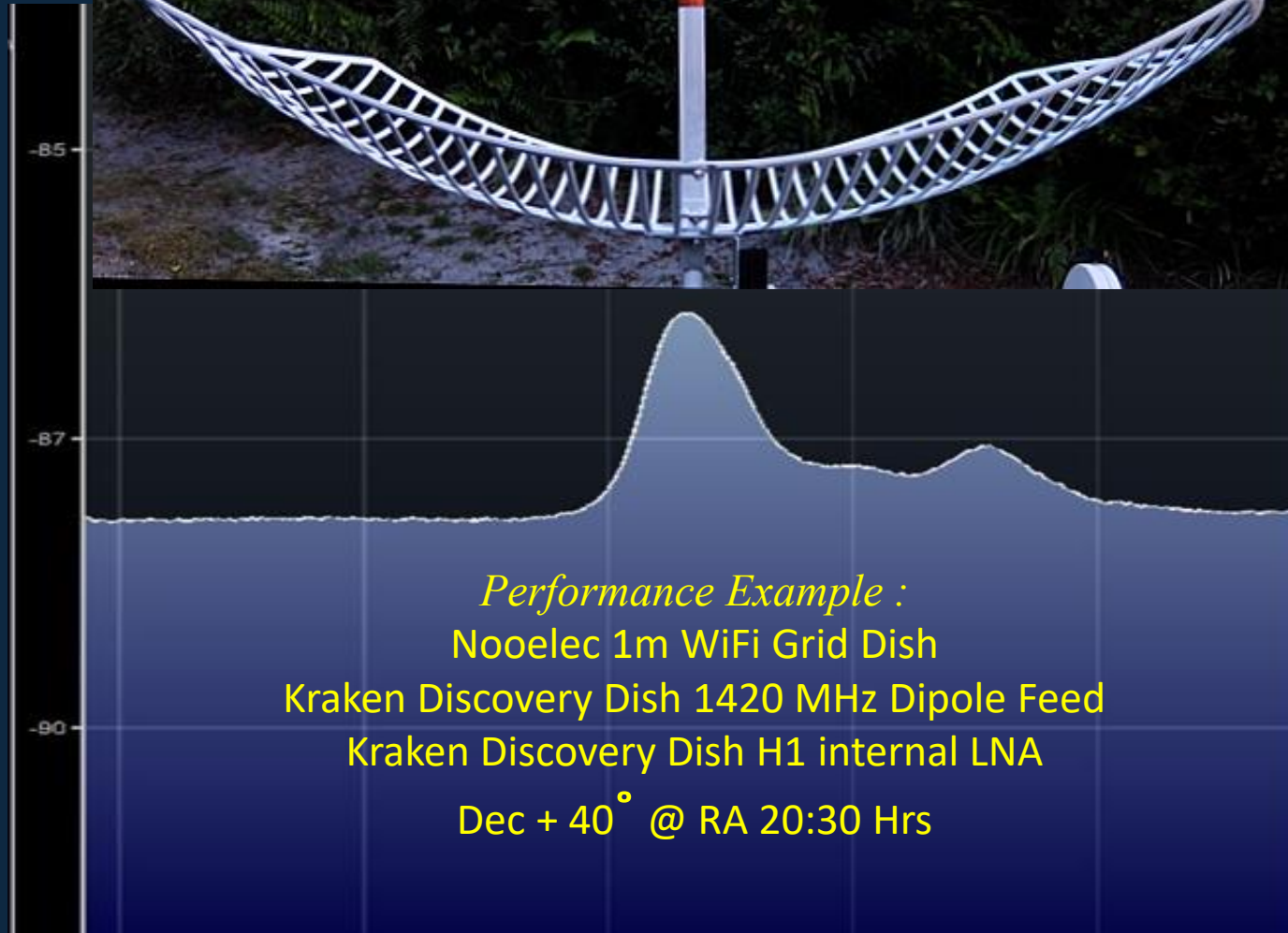
# Kraken Discovery Dish 1420 MHz Feed Evaluation

## IF\_avg Background Correction

**Materials Evaluation : Aluminum Foil works well for this application ( as does a Conductive Foam Box )**

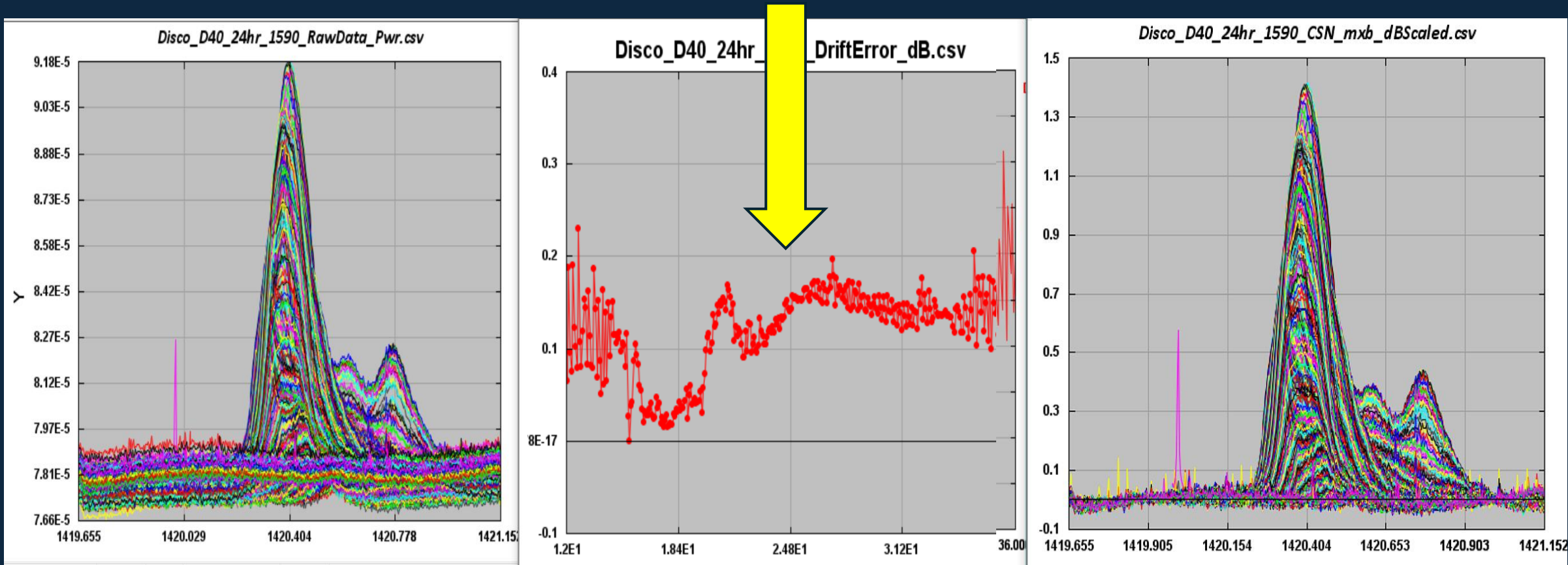


## *Performance Example*



Noelec 1m WiFi Grid Dish  
**Kraken** Discovery Dish 1420 MHz Dipole Feed  
**Kraken** Discovery Dish H1 internal LNA  
Dec + 40° @ 24 hour Drift Scan

Good Combination : High Quality Data & Low Frame-Frame Drift Error  
( this data set has the lowest drift error I've ever measured from a 24 hour scan  
with a small dish H Line Radio Telescope )

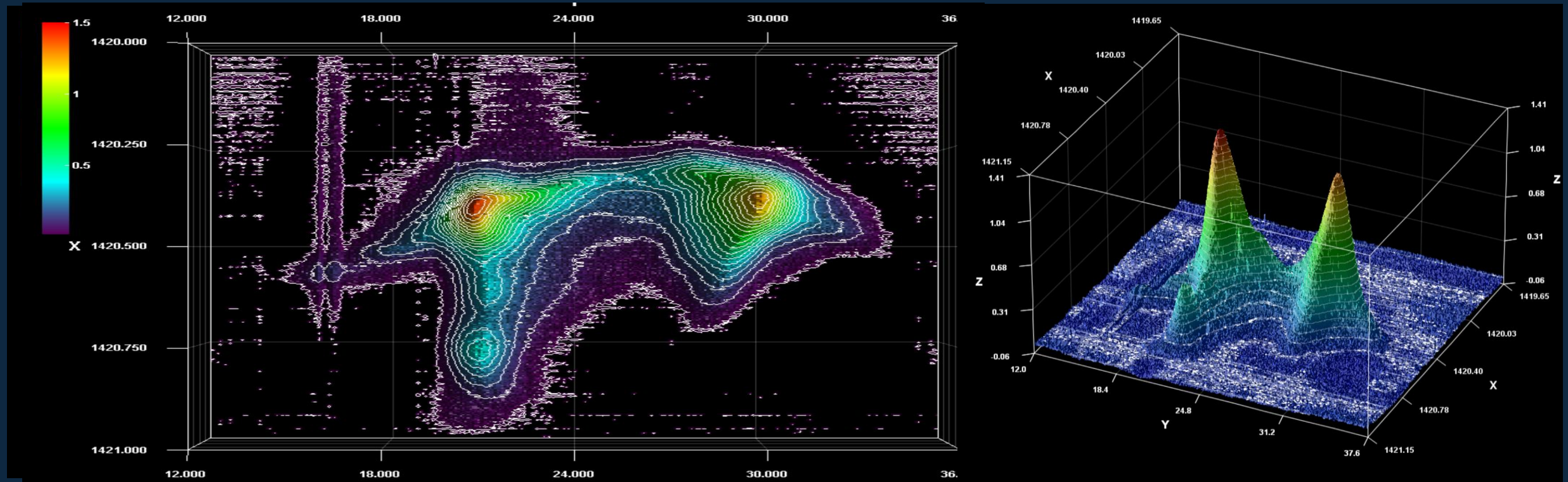


Nooelec 1m WiFi Grid Dish  
*Kraken* Discovery Dish 1420 MHz Dipole Feed  
*Kraken* Discovery Dish H1 internal LNA  
Dec + 40° 24 hour Drift Scan

Excellent **HIGH SIGNAL** / LOW NOISE Performance

Radio Telescope Performance Evaluated Using :

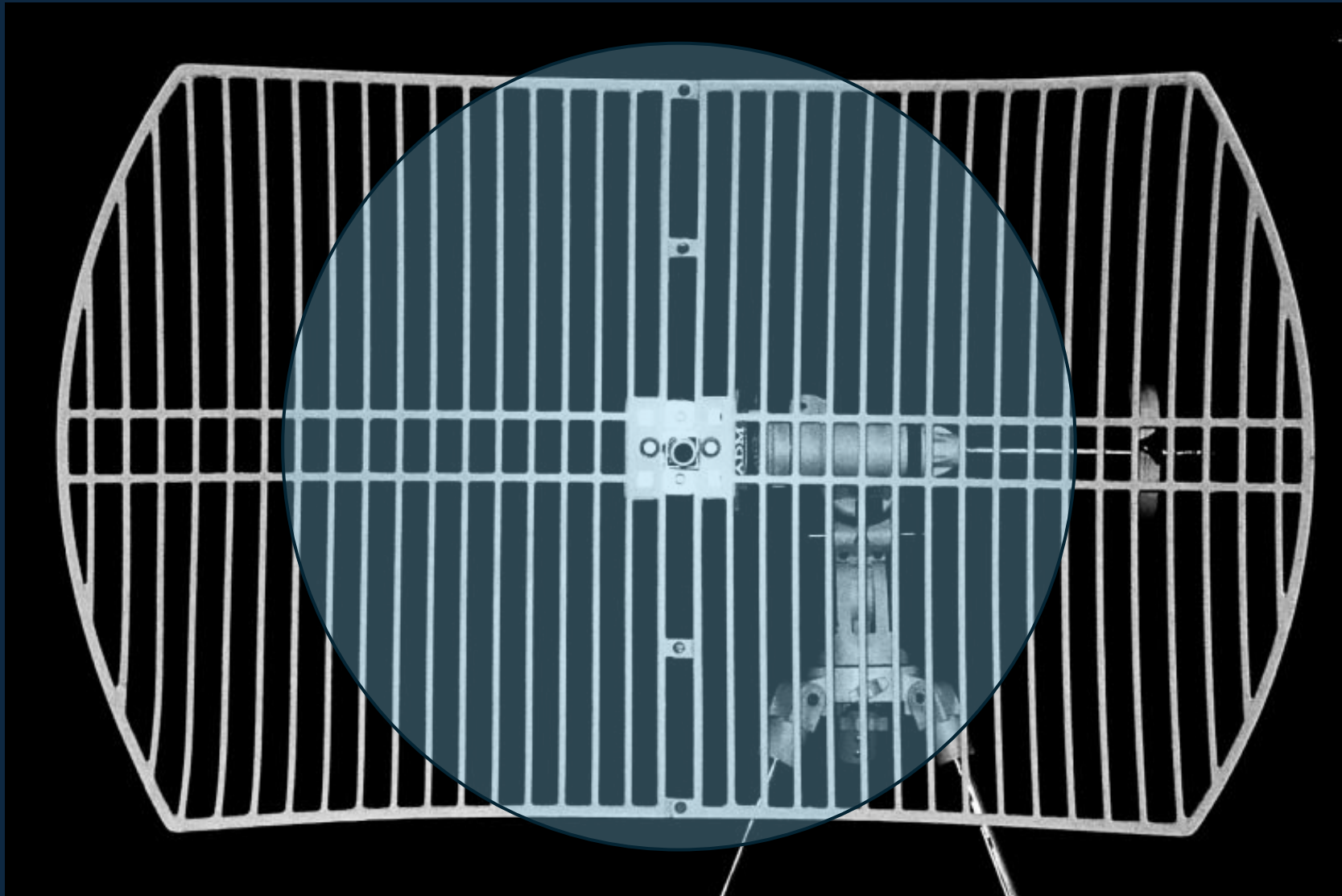
*AirSpy SDR# Studio* > *Kaminski IF\_average* > *Pettit HLine3D* > *Rinearn Graphics*  
<https://github.com/AP-HLine-3D/HLine3D>



## *Kraken Discovery Feed 1420 MHz Dish Options*

*1m x 0.6m WiFi Dish used for This evaluation*

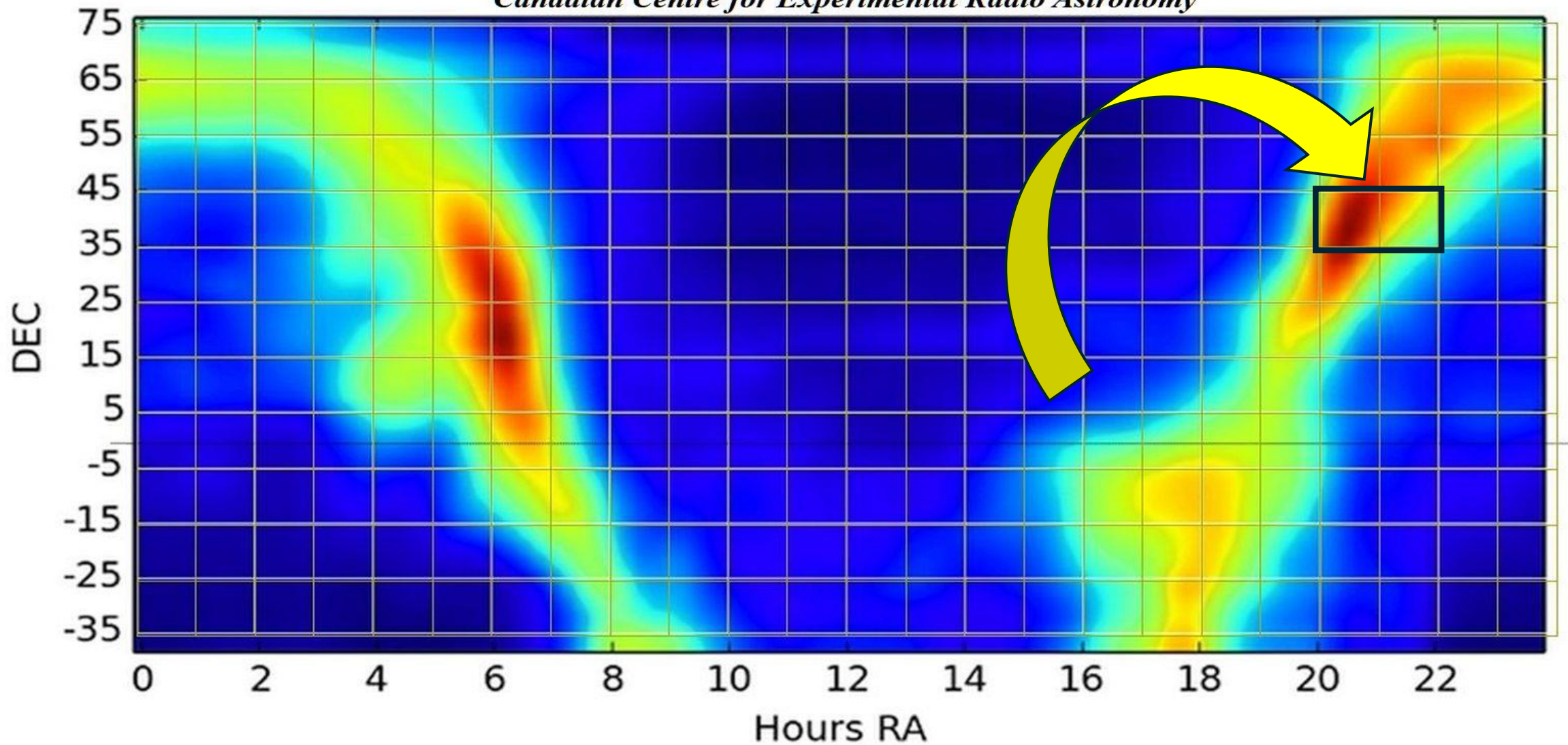
*0.65m Kraken Discovery Dish will be eval'd in the future*



*Drift Scan 2 Hr Scan @ Declination + 40° RA 20 - 22 Hrs*

CCERA

*Canadian Centre for Experimental Radio Astronomy*



## Drift Scan 2 Hr Scan @ Declination +40° RA 20 - 22 Hrs

Disco\_D40\_2hrs\_CSN\_mxb\_dBScaled.csv

